

# MATH Semester Examination 2023

Mathematics (MATH) · Semester Examination · 2023

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| Max. Marks | 75 | Time | 3 hours |
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Answer ALL questions from Part A and any FIVE from Part B.

## Part A (25 marks)

1. What is meant by Rank of a Matrix? Discuss its advantages and limitations. (5 marks)
2. Discuss the role of Mean Value Theorems in Differential Calculus. (5 marks)
3. Write short notes on Conditional Probability. (5 marks)
4. Illustrate Propositional Logic with a worked example. (5 marks)
5. Discuss the role of Permutations and Combinations in Graph Theory & Combinatorics. (5 marks)

## Part B

1. Compare Rank of a Matrix with Quadratic Forms. When would you prefer one over the other? (Unit 1)
2. Discuss the role of Systems of Linear Equations in Matrices & Linear Algebra. (Unit 1)
3. Define Mean Value Theorems. Derive or justify the key properties involved. (Unit 2)
4. Compare Taylor and Maclaurin Series with Jacobians. When would you prefer one over the other? (Unit 2)
5. Discuss the role of Conditional Probability in Probability & Statistics. (Unit 3)
6. Write short notes on Bayes' Theorem. (Unit 3)
7. Explain Propositional Logic with a neat diagram and a suitable example. (Unit 4)
8. Explain Predicate Logic with a neat diagram and a suitable example. (Unit 4)
9. Illustrate Permutations and Combinations with a worked example. (Unit 5)
10. Illustrate Pigeonhole Principle with a worked example. (Unit 5)